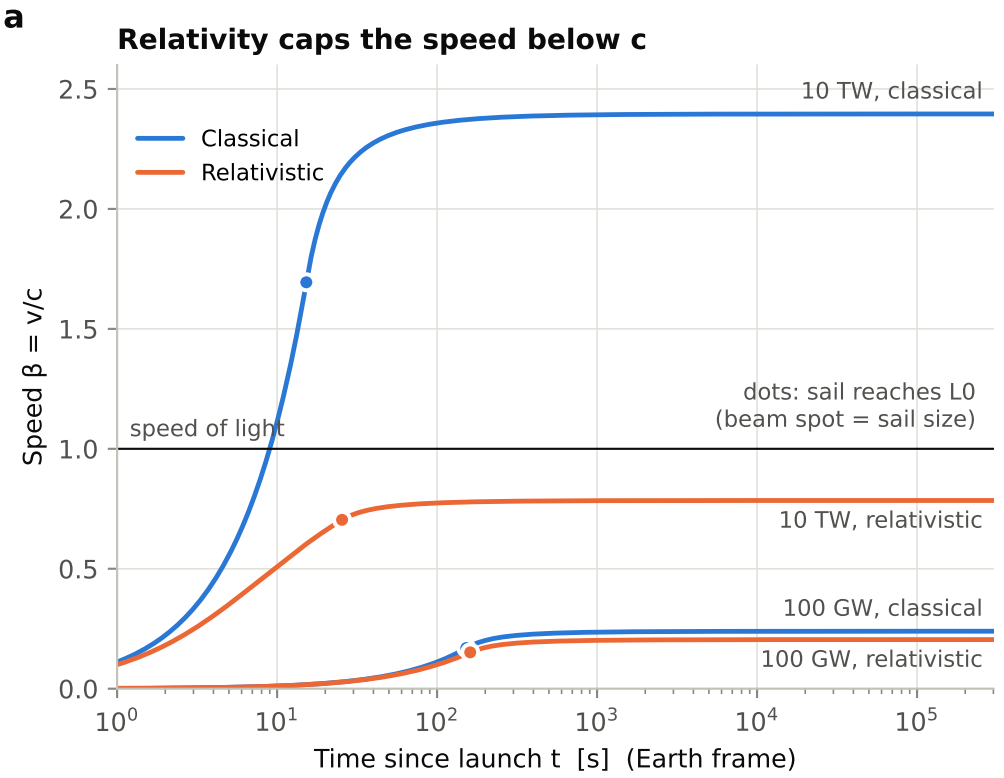
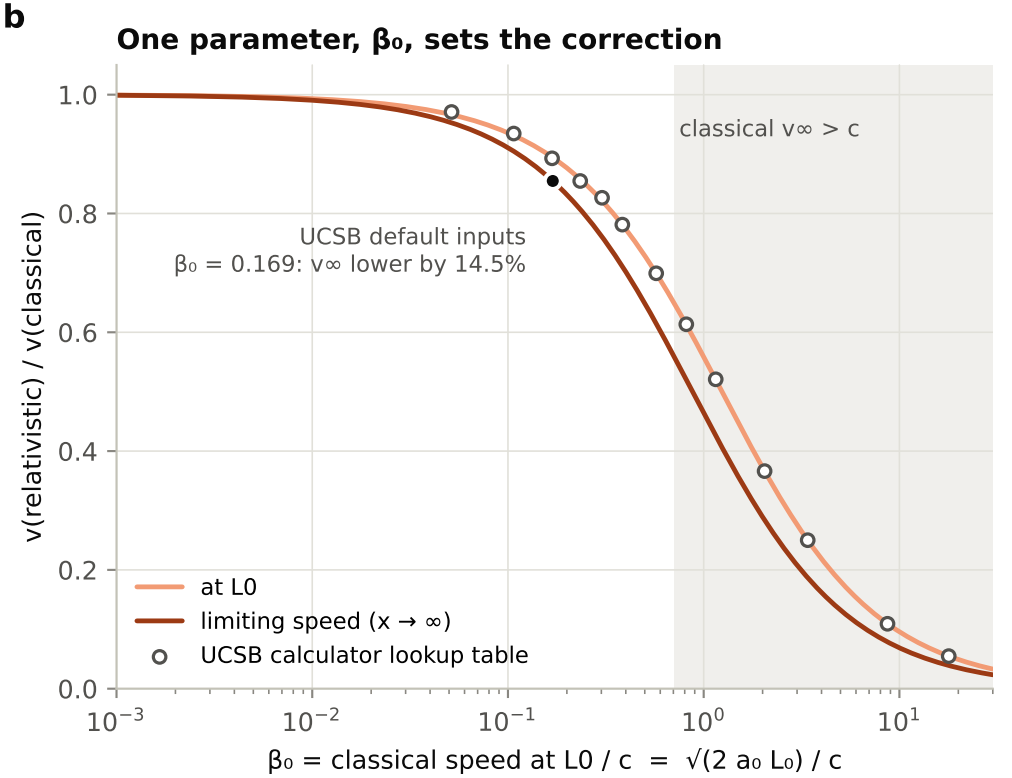

a
Relativity caps the speed below c
Classical
Relativistic
10 TW, classical
dots: sail reaches L0
(beam spot = sail size)
speed of light
10 TW, relativistic
100 GW, classical
100 GW, relativistic
Speed β = v/c
Time since launch t [s] (Earth frame)


b
One parameter, β₀, sets the correction
classical v∞ > c
UCSB default inputs
β₀ = 0.169: v∞ lower by 14.5%
at L0
limiting speed (x → ∞)
UCSB calculator lookup table
v(relativistic) / v(classical)
β₀ = classical speed at L0 / c = √(2 a₀ L₀) / c